

Flexible WSN Data Gathering through Energy-aware Adaptive Sensing

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Abstract—The multitude of Wireless Sensor Networks (WSNs) environments, being typically resource-constrained, clearly benefit from properties such as adaptiveness and energy-awareness, in particular, in presence of demanding data gathering applications. This paper proposes a self-adaptive, energy-aware sensing scheme for WSNs (e-LiteSense), which aims at self-adjusting the data gathering process to each specific WSN context, capturing accurately the behaviour of physical parameters of interest yet reducing the sensing overhead. The adaptive scheme relies on a set of low-complexity rules capable of auto-regulate the sensing frequency according to the parameters variability and energy levels. The proof-of-concept resorts to real-world datasets to provide evidence of e-LiteSense ability to optimise the data gathering process according to energy levels, improving the trade-off between accuracy and WSN lifetime.

I. INTRODUCTION

Wireless Sensor Networks (WSNs) have earned much attention in recent years due to research advances in the field of computation, robotics, microelectronics, nanotechnology and wireless communications, making possible the development of multifunctional low-cost wireless sensors, mostly fuelled by the Internet of Things (IoT) context [1]–[3].

The global performance and lifetime of this type of networks are typically dependent on the applications being supported and on the constraints affecting the infrastructure resources. The energy consumption is a critical aspect in WSNs as sensor nodes must optimally waste the least amount of energy possible to perform their tasks in order to maximise their lifetime. Research on this topic has identified the communication subsystem as the most demanding regarding to energy consumption [4]. This is true even for devices capable of energy harvesting. Other subsystems impacting on energy are sensing and processing.

The typical approach used to reduce energy consumption consists in aggregating data acquired regularly in order to reduce transmission events. However, the sensing subsystem also impacts on energy consumption, as several works have indicated that acquiring and processing data could even overcome the communication subsystem for some operational scenarios [5] [6]. This suggests that data should only be acquired when necessary, avoiding redundant information, even in presence of data aggregation.

In this context, the main objective of this work is to enhance the scope and efficiency of WSNs data gathering processes,

through an energy-aware adaptive sensing scheme able to balance accuracy in data acquisition with energy conservation. The proposed e-LiteSense scheme is self-adjustable, being able to adapt sensing to specific WSN data gathering requirements, while extending sensor's lifetime and, consequently, the network utility. Through low-complexity rules, devised to optimise the *sensing, processing, communicating* vectors (see Figure 1), e-LiteSense adjusts the sensing frequency based on the variation in scalar physical quantities being measured and the device's residual battery level. The obtained results, using real datasets, evince the effectiveness of the proposed rules in distinct WSNs scenarios.

This paper is organised as follows: related work is discussed in Section II; the sensing scheme design goals and rationale are described in Section III; the proof-of-concept and the corresponding evaluation results are discussed in Section IV; and the conclusions are summarised in Section V.

II. RELATED WORK

The definition of WSNs lifetime, although intrinsically related with battery lifetime, may vary depending on the application's functionality and behaviour, on the objective function and on the network topology in use. A comprehensive overview of the most relevant WSN lifetime maximisation techniques in the literature is provided in [7], where the underlying optimisation algorithms are classified as resource allocation, opportunistic transmissions, sleep-wake scheduling, routing, clustering, mobile relays and sinks, coverage and connectivity, optimal deployment, data gathering, network coding, data correlation, energy harvesting and beamforming, depending on their focus.

In particular, data correlation techniques take advantage of temporal-spatial data correlation characteristics to reduce the overall transmitted traffic and, by reducing/removing redundancy, they benefit energy conservation and network lifetime. When these techniques are combined with self-adaptive sensing schemes, data gathering and transmission may be substantially reduced, improving the versatility and lifetime of WSNs in heterogeneous environments.

Several adaptive sensing approaches have been proposed for WSNs taking into account the inherent computational and energetic limitations of these networks. In [8], an optimal sensing scheduling policy is proposed for a power harvesting system

equipped with a finite battery. The goal is to select sensing periods strategically so that the average sensing performance is optimised. In that system, the authors assume that performance depends on time duration between two consecutive sensing periods, and develop an algorithm where the sensing rates depend on the battery power levels of sensor nodes. However, this analytical study only considers the battery level of sensors disregarding the variability of the observed parameter.

A common approach aiming to reduce the network data communication load resorts to compressive sensing. It consists in obtaining information that is based on signal sparsity, statistical analysis and other related theory [9]. However, when necessary, the data decompression processes in sensor nodes are, in general, computationally inefficient [10]. In addition, the sparsity of the sensor data may change significantly due to their time varying nature, affecting the recovery of the measured signal [11].

In [12], an analytical study for sensing, compression and recovering signals from large datasets is carried out. The analysis resorts to probabilistic relations among variables involved in those processes to capture the space and time correlations of the signals of interest. The proposal lowers the number of transmissions and, consequently, the power consumption in sensor nodes. However, this study is mainly focused on reducing the transmission rate, disregarding the accuracy of the sensing process.

In fact, the complex relation between data analysis and data acquisition in adaptive sensing paradigms can be extremely powerful, as it allows a reliable estimation and signal detection in situations where non-adaptive detection fails [13]. For this reason, [13] investigates the signals' estimation over the adaptive detection problem. The authors present a general estimation procedure which may be satisfactory for a variety of cases, however, the authors recognise that the prior knowledge of some parameters might not be available in a real-life setting.

In [14], the authors establish a framework for collecting data from a WSN based on adaptive compression sensing (as in [12]), aiming to reduce the power consumption and the amount of information in sensing data. The paper proposes an algorithm for obtaining a more precise approximation of the measurements using the lowest possible energy. This proposal is also confined to adapt the transmission rate.

In [6], the authors worked in a prediction model for estimating battery lifetime. The methodology followed relied on the decomposition of the hardware operations by time periods, using datasheets of sensor nodes as reference.

In [5], the authors propose a system to calculate in runtime energy consumption of WSN platforms. The authors based the power consumption evaluation on separate pieces of hardware such as leds, data transmission, transmission power, micro-controller and CPU instructions.

The above review shows that most of the studies on adaptive sensing either: (i) assume a theoretical approach limiting their applicability to real WSN scenarios; or (ii) only cover a partial set of variables to optimise, such as the transmission rate, disregarding accuracy. This evinces the need of studying and

improving the trade-off between sensing events and measurements accuracy for distinct WSN contexts.

The present proposal extends a previous work by the same authors [15] in the sense that data acquisition rate is now, not only dependent on the observed variation in the scalar physical quantities measured, but also on the perceived devices residual battery level over time.

III. ENERGY-AWARE ADAPTIVE SENSING

This section identifies the main goals driving the design of the energy-aware adaptive proposal, followed by a detailed explanation of the underlying algorithms.

A. e-LiteSense architecture

This section includes an initial discussion clarifying how self-adaptive sensing may fit within WSN context. The envisioned architecture integrating e-LiteSense sensor nodes comprises three planes, as presented in Figure 1, and detailed below:

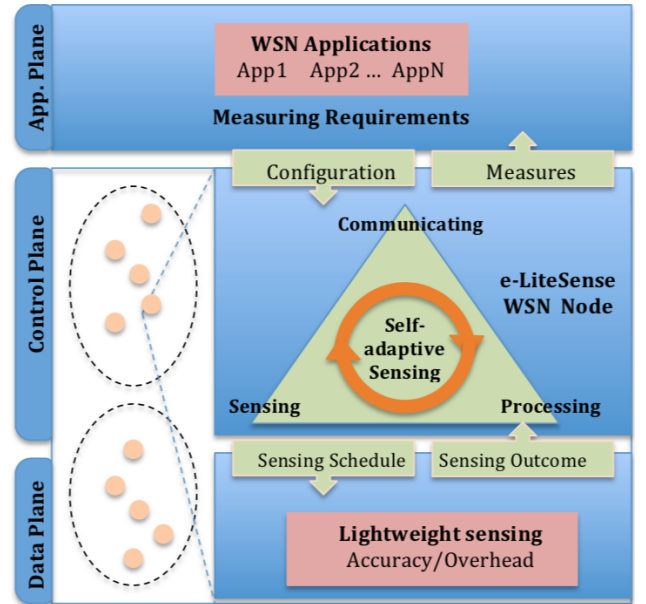


Fig. 1. e-LiteSense architecture

Application plane: WSNs sustain a plethora of application areas broadly categorised as: environmental monitoring; transportation and logistics; medical and health; security and emergencies; smart world; entertainment; retail, industrial control and automation; and military related applications [7]. Based on the requirements of each application context, specific measurement needs determine the configuration of sensor nodes. The obtained measures allow to keep track of crucial aspects of WSNs operation, such as the temporal evolution of the parameters under evaluation. The application plane is also responsible for processing measurement results reported by the control plane and for providing a visualisation component, when applicable. The required processing may involve results from single or multiple sensor measurements.

Control plane: The control plane, deployed in WSN sensor nodes, implements the e-LiteSense strategy. As detailed below, e-LiteSense allows an autonomous adjustment of sensing events, through lightweight algorithms which consider the main vectors involved in energy consumption: sensing; processing; and communicating. These self-adaptive algorithms are devised to obtain accurate measures of the WSN parameters under observation while reducing sensing overhead. Considering this trade-off, sensing schedule is adaptively configured, according to short/medium-term variations in the measured parameters. Conversely, the sensing outcome received from the data plane is processed (e.g., aggregated) to fulfil the application measurement needs.

Note that, the control plane may be deployed in sensor nodes directly (as represented in Figure 1) or in external coordination entities, according to SDN and Edge Computing principles or architectures. In this case, depending on the application measurement requirements, an SDN controller implementing e-LiteSense will interact with edge nodes through southbound interfaces, being the edge nodes responsible for configuring the WSN sensor devices according to e-LiteSense algorithms behaviour.

Data plane: At data plane, the measures from sensors are obtained following the defined schedule and passed to the control plane. The use of a flexible sensing scheme will allow to configure sensing events according to the application measurement purpose and dynamics.

B. e-LiteSense sampling algorithms

Ground truth shows that, independently of the application type, a high volume of data acquired leads to high accuracy in measurements. However, this also implies a high-energy consumption, not only for sensing but also for processing and transmitting subsystems.

Attending to the interrelation among the above subsystems, the e-LiteSense self-adaptive sensing scheme uses the variation observed in scalar physical quantities being measured and the residual battery level of the device in order to adjust the sensing frequency targeting the best trade-off between accuracy and sensor lifetime.

Basically, when the data being acquired does not vary significantly, the time interval until the next sensing event (ΔT_{next}) is increased, reducing its frequency and, consequently, the overall energy consumed. This needs to be accomplished without noticeable impact on measurements accuracy. Conversely, if a significant variation is observed in the sensed values, the time scheduling for the next sensing event is decreased aiming to improve the ability in identifying and capturing the variable fluctuation. In both scenarios, the rules defining the time interval until the next sensing event are defined considering the current power level of the sensor's battery.

The proposed sensing scheme, relying on simple arithmetic processes, was devised having low overhead, simplicity of implementation and low resource consumption as main design goals. The algorithms 1 and 2 illustrate the expected behaviour

through a high-level code description, and Table I describes their main elements.

Inspired on TCP RTT estimation mechanism [16], defining the parameter scalar variation resorts to the moving average of observed values ($\bar{X}_i$) and mean deviation ($\bar{V}_i$) as a simple way to compute an approximation for the standard deviation (see lines 6 and 7 in Algorithm 1).

On this basis, after a sensing event S , the algorithm compares how the measures evolve ($\bar{X}_{i-1} \xleftarrow{\bar{V}_{i-1}} \bar{X}_i \xleftarrow{\bar{V}_i} S$) in order to define the time interval until the next event (ΔT_{next}). If the current variation ($\bar{V}_i$) is higher than the last variation ($\bar{V}_{i-1}$), the measured value is changing significantly, so the time interval to the next sensing event is reduced aiming at getting an accurate view on the new pattern. Conversely, if the current variation is lower than the previous one, the observed physical quantity is stable, which allows using a higher interval until the next sensing event, conserving overall device's energy (Algorithm 2 performs this comparison and returns the corresponding adjustment value).

```

main()
1   $\bar{X}_{i-1} \leftarrow 0$ 
2   $\bar{V}_{i-1} \leftarrow 0$ 
3  repeat
4     $S \leftarrow \text{getSense}()$ 
5     $\Delta T_i \leftarrow \text{timestamp of } S$ 
6     $\bar{X}_i \leftarrow (1 - \alpha)\bar{X}_{i-1} + \alpha S$ 
7     $\bar{V}_i \leftarrow (1 - \beta)\bar{V}_{i-1} + \beta|\bar{X}_i - S|$ 
8     $\Delta T_{next} \leftarrow \Delta T_i + (\Delta T_i \times \text{getScale}(\bar{V}_i, \bar{V}_{i-1}))$ 
9     $\Delta T_{next} \leftarrow \max(\min(\Delta T_{max}, \Delta T_{next}), \Delta T_{min})$ 
10    $\bar{X}_{i-1} \leftarrow \bar{X}_i$ 
11    $\bar{V}_{i-1} \leftarrow \bar{V}_i$ 
12   wait( $\Delta T_{next}$ )
13 until end execution;
```

Algorithm 1: Setting ΔT_{next}

```

getScale( $\bar{V}_i, \bar{V}_{i-1}$ )
1 if ( $\bar{V}_i > \bar{V}_{i-1}$ ) then
2   return  $-((\text{getBattery}()) \times (\epsilon_{max} - \epsilon_{min}))/100 + \epsilon_{min}$ 
3 else
4   return  $((100 - \text{getBattery}()) \times (\epsilon_{max} - \epsilon_{min}))/100 + \epsilon_{min}$ 
```

Algorithm 2: Defining scheme reactivity

The algorithm reactivity (*i.e.*, ΔT changing scale) is defined according to the residual energy in node's battery. The main rational relies on: (i) a fast reduce and slow increase of ΔT in presence of high energy levels and (ii) a slow reduce and fast increase of ΔT in presence of low energy levels. Such reactivity is bounded by limit values (ϵ_{min} and ϵ_{max}) adjusted according to the application and type of measured phenomena. Algorithm 2 presents such operations.

An additional constraint is used to prevent ΔT from growing indefinitely (ΔT_{max}), thus guaranteeing a minimum number of samples per time unit. Conversely, the maximum frequency of sampling (ΔT_{min}) is also limited so that the interval between sensing events does not tend to zero, which

TABLE I
DESCRIPTION OF ALGORITHM 1 AND 2 ELEMENTS

$\bar{X}_i$	- moving average of the observed parameter.
$\bar{X}_{i-1}$	- moving average calculated in the previous event.
S	- scalar value observed in the current sensing event.
α	- weight of S in the moving average $\bar{X}_i$.
$\bar{V}_i$	- current mean deviation of observed values.
$\bar{V}_{i-1}$	- mean deviation in the last sensing event.
β	- the weight of the current deviation $ \bar{X}_i - S $.
ΔT_{next}	- the time interval until the next sensing event.
$\Delta T_{min} \Delta T_{max}$	- upper and lower ΔT bounds.
$getScale()$	- function ruling the scheme's reactivity.
$\epsilon_{min} \epsilon_{max}$	- scale reactivity boundaries.

would result in an overwhelming resource consumption. They might be set according to the application scenario.

Although not represented in Algorithms 1 and 2, e-LiteSense also provides a critical operation mode which is triggered if a threshold value is observed in the measurements. In this operation mode, sensing events are performed at the highest frequency regardless the sensor's battery level. Monitoring human vital signs or sensitive environments are examples in which such feature is essential, even though redundant batteries are in place.

IV. EVALUATION RESULTS

This section presents the proof-of-concept for e-LiteSense scheme by assessing its ability in reducing energy consumption while providing accurate measurements about the observed phenomena. In addition, the analysis addresses the impact of different algorithm settings on measurement accuracy and energy burden, in particular, the impact of reactivity (ϵ_{min} and ϵ_{max}) and time boundaries (ΔT_{min} and ΔT_{max}).

A. Methodology of tests

In order to evaluate the performance of e-LiteSense, a software-based prototype built according to TelosB motes' specifications [5] was implemented, allowing to assess a fine-grain power profile of the main subsystems (*i.e.*, communication, sensing and processing) of commercial sensors running the proposed scheme.

Measurements accuracy and energy consumption are then evaluated resorting to real datasets previously collected and publicly available^{1,2}. These datasets consist of: (i) an outdoor sensor measuring environmental temperature along six hours; (ii) an outdoor sensor measuring environmental humidity along six hours; and (iii) an indoor sensor measuring environmental temperature along four years, located in Luxembourg City.

In the first two test scenarios, the sensors were exposed to a steam of hot water to spark humidity and temperature levels in order to assess e-LiteSense's accuracy in identifying sudden changes in the observed parameters.

The accuracy evaluation is performed through the estimation of the *correlation coefficient* and the *Mean Squared Error* (MSE) between the measured values and the real phenomena

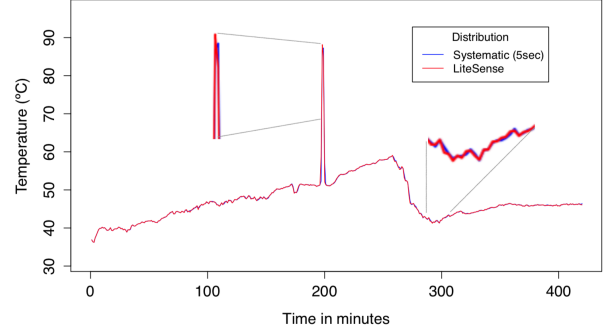


Fig. 2. Measurements for the outdoor temperature sensor

TABLE II
MEASUREMENT ACCURACY FOR DIFFERENT SCENARIOS

Luxembourg		Temperature		Humidity	
Correlation	Error	Correlation	Error	Correlation	Error
0.99669	0.00653	0.99968	0.02078	0.99992	0.00142

dynamics. These metrics are commonly used to evaluate the accuracy of sampling-based estimators.

The parameterisation of tests, discussed in Section IV-B, considers $\Delta T_{min} = 5$ and $\Delta T_{max} = 45$, meaning that, at least, one sensing event occurs every 45 seconds for scenarios (i) and (ii) and every 45 minutes for scenario (iii). The reactivity factor is defined as $\epsilon_{min} = 0.05$ and $\epsilon_{max} = 0.30$, meaning that ΔT_{next} will vary between 5% and 30% in each cycle. The overall impact of adjusting these parameters is further discussed in Section IV-C. In addition, the weight of the last observations defining $\bar{X}_{i-1}$ and $\bar{V}_{i-1}$ are $\alpha = 0.7$ and $\beta = 0.7$, respectively.

B. Evaluating e-LiteSense performance

Regarding the ability of producing accurate measurements in diverse application scenarios, Figure 2 presents the distribution of sensed values along the tests for the temperature sensor. The almost complete overlap of e-LiteSense and original dataset (systematically sensed every 5 seconds), even when the data series exhibit sudden variability, is corroborated by the high correlation coefficient and low MSE presented in Table II. It is important to highlight that such high accuracy is achieved reducing the number of sensing events in 54% (from 5041 in the original dataset to 2800) through e-LiteSense. Similar performance is achieved for all datasets, which demonstrates the effectiveness of e-LiteSense in self-adjusting the sensing frequency accordingly towards the observed parameter dynamics.

Designed with the aim of extending the sensor's lifetime compared to the predominantly deployed scheme, *i.e.*, systematic sensing, the self-adaptiveness nature of e-LiteSense also achieves high efficiency while optimising energy consumption. As illustrated in Figure 3, for the Luxembourg scenario, e-LiteSense has extended the sensor operational lifetime in

¹http://issnip.unimelb.edu.au/research_program/download

²<https://doi.org/10.4121/uuid:284697ee-9dbe-4f63-b4fb-d5e8e3bc4606>

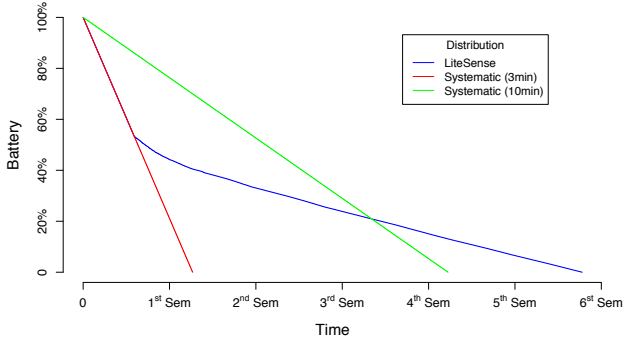


Fig. 3. Power consumption for Luxembourg sensor

approximately 5 times when compared with the systematic approach currently under operation in this environment, namely 3 minutes. Even compared with a more conservative systematic scenario, *i.e.*, one sensing event every 10 minutes, a sensor running an e-LiteSense instance would extend its operation in around 1 year after a sensor with systematic sensing completely depleted its battery.

As detailed in Section III-B, e-LiteSense adjusts the reactivity of the sensing frequency according to the sensor's residual battery level. In this way, Figure 4 presents the impact of such behaviour on the measurements accuracy along the battery consumption stages. It represents the dispersion of temperature per minute measured through the proposed scheme in comparison to the original dataset for the Luxembourg scenario. According to this analysis, as the battery is depleted, measurement error increases. Therefore, balancing the extension of the sensor lifetime with the tolerated error for a specific application is relevant for configuring the proposed scheme, *i.e.*, reactivity and time boundaries, discussed in Section IV-C.

C. Analysing the algorithm reactivity and time boundaries

Following the design goals discussed in Section III, e-LiteSense scheme provides the flexibility of adjusting the algorithm reactivity and ΔT boundaries in order to foster its deployment in multiple scenarios and applications. This parameterization might be defined beforehand for a specific scenario or adjusted dynamically by an external controller (in response to the underlying application requirements or to the values collected in different sensors).

In this way, Table III presents the impact of adjusting the scheme reactivity, *i.e.*, ϵ_{min} and ϵ_{max} for all datasets. These tests consider $\Delta T_{min} = 5$ and $\Delta T_{min} = 45$ as fixed values. As observed, although varying slightly, the overall accuracy is kept high for the considered scenarios. In these cases, the major impact observed is the energy consumed, which vary approximately 7% for all scenarios, in average.

Similarly, Table IV presents the impact of adjusting e-LiteSense's time boundaries on measurement accuracy and energy consumption. In this case, the most conservative scenario, (*i.e.*, $\Delta T_{min} = 5$ and $\Delta T_{max} = 60$) means that, at least

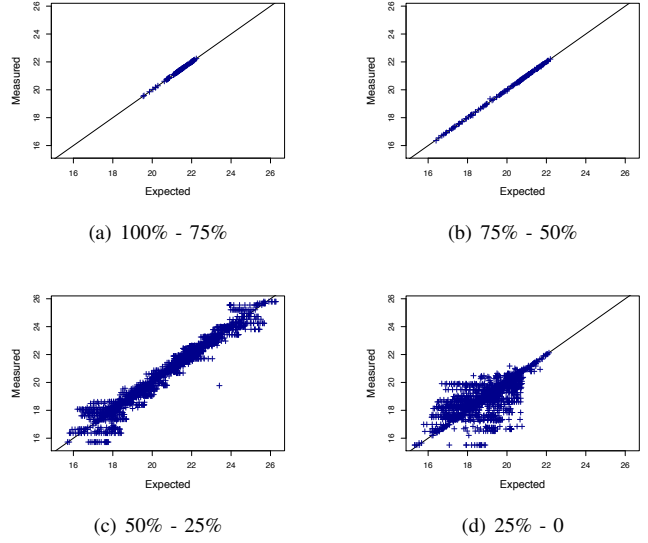


Fig. 4. Dispersion of observed values for distinct battery levels

TABLE III
IMPACT OF ADJUSTING THE ALGORITHM REACTIVITY

$\epsilon_{min}:\epsilon_{max}$	Luxembourg			Temperature			Humidity		
	Correlation	Error	Battery	Correlation	Error	Battery	Correlation	Error	Battery
0.05:0.3	0.998	0.011	78%	0.999	0.025	55%	0.999	0.003	54%
0.05:0.5	0.999	0.005	79%	0.999	0.060	57%	0.999	0.002	58%
0.05:0.7	0.999	0.003	81%	0.998	0.085	59%	0.999	0.003	59%
0.05:1.0	0.999	0.002	84%	0.999	0.020	61%	0.999	0.001	62%

one sensing event occurs every minute³. On the other hand, the most eager scenario, (*i.e.*, $\Delta T_{min} = 5$ and $\Delta T_{max} = 15$), means that, at least four sensing events occur every minute. For all setups, the maximum frequency is eleven sensing events per minute and the reactivity is configured as $\epsilon_{min} = 0.05$ and $\epsilon_{max} = 0.3$

As observed for the reactivity adjustments, in the considered scenarios, tuning the time boundaries does not affect the high accuracy of measurements significantly. Again, the energy consumption is the main affected aspect, varying around 20% for the temperature and humidity datasets.

TABLE IV
THE IMPACT OF ADJUSTING TIME BOUNDARIES

$\Delta T_{min}:\Delta T_{max}$	Luxembourg			Temperature			Humidity		
	Correlation	Error	Battery	Correlation	Error	Battery	Correlation	Error	Battery
5:15	0.999	0.0001	100%	0.999	0.010	71%	0.999	0.0008	73%
5:30	0.999	0.0002	95%	0.999	0.012	62%	0.999	0.0009	61%
5:45	0.999	0.0009	81%	0.999	0.020	55%	0.999	0.001	57%
5:60	0.999	0.0032	74%	0.999	0.025	55%	0.999	0.003	54%

V. CONCLUSIONS

The deployment and operation of WSNs, very dependent on the type of applications and user services being supported, is commonly conditioned by the amount of available resources (energy, processing and transmission capacity, etc.). To mitigate these problems, particularly relevant for demanding data

³For the Luxembourg scenario, it means, at least, one sensing event every hour. The same time scale is applied for all tests in such dataset.

gathering processes, this paper has proposed e-LiteSense - a self-adaptive, energy-aware sensing scheme for WSNs. The present study evinces that this scheme is able to self-adjust data gathering to a specific WSN context, capturing accurately the behaviour of physical parameters of interest. Relying on low-complexity rules, e-LiteSense is able to auto-regulate the sensing frequency according to the parameters' variability and energy levels. Based on real datasets, the results showed that e-LiteSense was able to measure accurately the parameters under observation while reducing significantly the number of sensing events, evincing its ability to improve the trade-off between accuracy and WSN lifetime.

As future work, the architecture presented in Figure 1 will be extended in order to adopt edge computing facilities (e.g., using more powerful devices as head nodes) in order to adjust dynamically parameters such as ΔT_{min} , ΔT_{max} , ϵ_{min} and ϵ_{max} according to the particularities of the deployed scenario.

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